

L03 Individual Contribution Journal

Name: (Okafor, Ajmal, & Wilturner, 2026)

Course: ITAI 1371 – Machine Learning

Assignment: L03 Lab – ML Workflow & Learning Types

ARINZE What I Did on This Project:

- **Executed the Technical Notebook:**

I worked through all 25 cells of the Module_03_Lab_Exercise.ipynb notebook in Google Colab. I made sure every stage of the machine learning workflow was properly run—from initial data loading and cleaning to comparing supervised, unsupervised, and reinforcement learning methods. I added inline comments to break down complex sections like feature transformations and metric evaluations, then exported the notebook as a clean, readable PDF (L03_ArinzeOkafor_ITAI1371.pdf).

- **Managed Group Coordination:**

Aside from completing my technical tasks, I took the lead on keeping the team on schedule. I actively followed up with everyone to ensure they completed their individual parts and sent over their reflections and task breakdowns on time.

- **Document Assembly & Quality Check:**

Once everyone submitted their work to me, I gathered all team inputs, double-checked the formatting, and compiled everything into a structured submission package. I verified all file names against the class guidelines (L03_YourGroupName_ITAI1371.pdf, L03Journal_R_YourGroupName_ITAI1371.pdf, and

L03Journal_C_YourGroupName_ITAI1371.pdf) so everything was ready for Canvas before the deadline.

Reflection on the Process

Taking on both the technical execution and the team synthesis gave me a complete view of how machine learning projects work in practice. Getting the code to run correctly is only half the effort—keeping communication clear and organizing everyone's work into a solid final deliverable is just as critical.

Hamza Ajmal

Contribution Journal

Team structure. I completed this lab on my own, so all the work described below is mine. There were no teammates to divide tasks with.

Running the notebook. I uploaded Module_03_Lab_Exercise.ipynb to Google Colab and ran it from top to bottom, checking the output of each stage before moving on. I confirmed the Wine dataset had 178 samples, 13 features and three classes (59, 71 and 48 samples), with no missing values. I looked at the class distribution chart and the correlation heatmap, then followed the workflow through the train/test split (142 training and 36 test samples), training a Logistic Regression and a Decision Tree, and evaluating them. Logistic Regression scored 88.9% (32 of 36 correct) and the Decision Tree scored 83.3% (30 of 36). I read the classification reports and the confusion matrix to see how each model did on individual classes.

Task 1: feature selection. To understand how feature choice affects a model, I tested several alternative feature sets rather than accepting the first one. I chose flavonoids, color intensity and proline because their class averages differed clearly across the wine classes. That model reached 91.7% (33 of 36) compared with 88.9% for the original four features, and I recognized that this was a one-sample difference and should not be overstated. I wrote the explanation into a comment above my feature choice, using the class averages I looked up (for example, flavonoids of 2.98 in class_0 versus 0.78 in class_2). I also added two cells that print the class averages for the original four features and for my chosen set, so that the notebook shows the evidence behind my explanation. While doing this, I noticed that one table I was going to use described a different feature set from the one the models were trained on, and I corrected it.

Assessment cell. I ran the assessment cell, which scored 5/5, and I checked my understanding by explaining in my own words why the house-price scenario is supervised, the chess scenario is reinforcement learning, and the news-topics scenario is unsupervised.

Reflection cell and reflective journal. I completed the notebook's reflection cell with the analysis of my own results. I wrote the real-world application and key learnings sections myself, and I revised my wording after feedback. I also wrote the separate reflective journal, including its connection to Lab 2, and revised it after feedback.

Export. I ran all cells, exported the notebook to PDF, and named the files following the assignment convention.

Jonathon Wilturmer Individual Contribution Journal

For this lab, I worked through the whole notebook from start to finish. I ran all 25 cells in order and made sure each one actually finished and gave a real output before moving on, instead of just running everything at once and hoping for the best. I didn't just leave the default features the notebook had set up. I picked three different ones and re-ran the model to see how the accuracy changed compared to the original Logistic Regression result. That part took the most thinking, I had to go through the full list of 13 features and actually consider which ones made sense to test together, not just grab random ones.

I also wrote the reflection section explaining supervised, unsupervised, and reinforcement learning the way I understood them after working through the wine classification example. I also broke down why I think Logistic Regression beat the Decision Tree, and what I'd try next if I wanted to push the accuracy higher.

Once everything was filled in, I ran the whole notebook again top to bottom to make sure nothing broke, then exported it to PDF and checked that every cell and output was readable before naming the file.

This lab took a few hours total, most of that time went into making sure I actually understood why things were happening instead of just clicking through to get it done.